From Paper to Proof

Setting up Lean · BGSMath, UPC, November 2026

You do **not** need a laptop in class: the sessions are demonstrations. You redo the demonstrations and the exercises at home. For sessions 1 and 2 a web browser is enough. For sessions 3 and 4, install Lean on your laptop.

1 In the browser (sessions 1 and 2)

1. Open `live.lean-lang.org`. Nothing to install.
2. Open a course file, for example `Exercises1.lean`, in the `BGSMath` folder of `github.com/mkaratarakis/from-paper-to-proof`. Click *Raw*, then copy all of it.
3. In the browser, delete what is in the editor and paste the file in.
4. Wait for the first line, `import Mathlib`, to load. The first time, this can take a minute or two.
5. The panel on the right is the *infoview*. Click on a line, and it shows what Lean knows there.

To keep your work, copy it back into a file before you close the page.

2 On your laptop (sessions 3 and 4)

Allow about 30 minutes, a good internet connection, and a few gigabytes of disk space. These are the steps from `lean-lang.org/install`, where you also find screenshots.

1. Install **VS Code**, from `code.visualstudio.com`.
2. In VS Code, open the Extensions panel (Ctrl+Shift+X; on a Mac, Cmd+Shift+X). Search for `lean4` and install **Lean 4**. A setup guide opens: follow it. It installs Lean.
3. Create a project with Mathlib: click the $\forall$ symbol at the top right of VS Code, then *New Project…* and *Create Project Using Mathlib…*. Choose a folder and a name, for example `bgsmath-lean`. The first time, Mathlib is downloaded: this takes a while.
4. Download the course files from the same address: *Code* $\rightarrow$ *Download ZIP*. Copy the `.lean` files of its `BGSMath` folder into the project folder.
5. **Test it.** Open `Test.lean` from the project folder. Click on the line `#eval 2 + 2`: the infoview shows `4`, and nothing is underlined in red.

If Lean starts "building" thousands of files, stop it: close VS Code, open a terminal in the project folder, and run `lake exe cache get`. Then open VS Code again.

3 An AI agent (session 3, optional)

- Session 3 shows a coding agent, Claude Code, writing Lean proofs. Other agents work in the same way.
- Install instructions: `code.claude.com/docs`. It needs an account, and most plans are paid.
- To let the agent see what you see in the infoview, add **lean-lsp-mcp**: follow `github.com/oOoooOo/lean-lsp-mcp`.
- No agent? Every exercise can also be done by hand.

4 If you are stuck

- Use the browser for now: it is enough for sessions 1 and 2.
- Ask at the start of the next session, or on the Lean Zulip chat, stream *#new members*: `leanprover.zulipchat.com`. Questions of every level are welcome.